Amogh Merudi

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Experience

Research and Development Intern, *Tree House Ventures, Remote / Toronto, ON* **Oct 2025 – Jan 2026**

Client – Artha Wallet, USA

- Engineered the AI Price Scrapper platform using Python, FastAPI, and web automation, resulting in dynamic, near real-time market price monitoring.

Client – Ory, USA

- Improved information retrieval precision by evaluating ReRAG, a sophisticated retrieval-augmented generation framework incorporating reinforcement learning and contextual re-ranking.

Director of Technology, *Ontario Youth Sustainability Solution, Toronto, ON* **Mar 2024 – Sept 2025**

- Led end-to-end development of the official OYSS website using React, TypeScript, and Tailwind CSS, delivering event pages, blog infrastructure, and a centralized podcast hub to support organization-wide campaigns.
- Designed and implemented backend integrations to enable dynamic content updates (events, blogs, podcasts), reducing manual site changes and improving content turnaround for non-technical team members.
- Improved website accessibility and performance by enforcing consistent component patterns and accessibility-aware UI design, ensuring the site was usable across devices and audiences.

Market Expansion & Admissions Research Associate, *GrowME.ai, Remote.* **Apr 2025 – Sept 2025**

- Supported GrowME's expansion from the US into the Canadian higher education market
- Researched and documented Canadian university admissions systems, including OUAC, program structures, and timelines
- Advised on key differences between US and Canadian application processes to localize student guidance
- Collaborated with a small team to translate admissions insights into actionable onboarding and advisory content

Co-Founder, President, *JFSS Photography Club, Toronto, ON* **Oct 2024 – May 2025**

- Founded and led a school-wide photography club fostering creative visual storytelling.
- Organized monthly workshops and curated the digital magazine Light of Ecstasy.
- Managed and motivated a team of 30 members, 10 executives and 20 general members, to successfully operate the JFSS Photography Club.

Projects

ERoute — Google "Build with AI" Track Winner, Hack Canada 2026 | *Next.js, TypeScript, Mapbox GL, Three.js, MongoDB, LLM APIs,*

- Built an AI-powered emergency routing platform that recommends optimal hospitals based on symptom severity, distance, congestion, and predicted wait times.
- Engineered an interactive geospatial dashboard using Mapbox GL to visualize hospital congestion, demand heatmaps, and emergency network coverage.
- Implemented a voice-based AI triage pipeline that classifies patient urgency and dynamically adjusts hospital routing decisions.
- Developed a 3D hospital infrastructure simulator with React Three Fiber enabling planners to model ER capacity and analyze patient redistribution across city networks.

PRISM – Pull Request Insight & Supervision Machine | *FastAPI, Node.js, React, TypeScript, Tailwind CSS, Gemini, GitHub Actions*

- Built a GitHub-integrated AI system that analyzes pull requests to explain change intent, surface risk, and summarize code behavior in plain English.
- Combined deterministic signals (PR size, files touched, lint results) with LLM-based semantic reasoning to flag high-risk changes in areas like authentication, infrastructure, and configuration.
- Designed a modular backend separating AI analysis, scoring, and synthesis to support multiple models and evolving review heuristics.
- Developed a dashboard to visualize repository health trends and recent PR risk signals over time.

Atari — Best Duo Hack Winner, FraserHacks 2024| *JavaScript, p5.js, HTML, CSS*

- Built an interactive asteroid-style arcade game using the p5.js library.
- Designed spaceship mechanics for rotation, movement, and laser firing with responsive keyboard controls.
- Implemented asteroid spawning, collision detection, level progression, and debris animation for realistic gameplay.
- Integrated real-time score tracking and restart logic for continuous play experience.

OYSS Website | *React, TypeScript, Tailwind CSS, HTML*

- Designed and built the official website for Ontario Youth Sustainability Solutions
- Developed modular React components for blogs, podcasts, and campaign pages with dynamic content rendering.
- Implemented responsive layouts and accessibility features aligned with OYSS's sustainability branding.

Education

University of Toronto **Sept 2025 – Present**
Bachelor of Science, Computer Science *Toronto, ON*

John Fraser Secondary School **Sept 2023 – June 2025**
High School Diploma, Ontario Scholars Certificate *Toronto, ON*

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, HTML, CSS
Frameworks/Libraries: React, Flask, Tailwind CSS, p5.js
Tools: Git, VS Code, PyCharm, Notion, Apple Developer Tools, Vercel